

Problem-Solution Fit

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] ☐ Students preparing for examinations ☐ College and university learners ☐ Job aspirants preparing for technical interviews ☐ Educators and trainers ☐ E-learning platforms and institutions	6. CUSTOMER LIMITATIONS ☐ Limited access to personalized concept evaluation tools ☐ Difficulty expressing concepts clearly during interviews ☐ Limited opportunities to receive immediate expert feedback ☐ Lack of awareness about conceptual weaknesses ☐ Time constraints during exam preparation	5. AVAILABLE SOLUTIONS • Multiple Choice Question (MCQ) Platforms • Online Learning Platforms • Teacher/Peer Feedback • Keyword-based Assessment Systems
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2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

Students often struggle to determine whether they truly understand a concept or have simply memorized it. Traditional assessments rely on keyword matching or objective questions and fail to identify misconceptions or missing concepts. This problem occurs frequently during exam preparation, interview practice, and self-learning sessions, especially whenever students revise technical concepts.

9. PROBLEM ROOT / CAUSE [RC]

Traditional assessment systems focus primarily on keyword matching and objective testing rather than evaluating actual conceptual understanding. Students often memorize answers instead of explaining concepts in their own words. Existing tools rarely analyze spoken explanations or identify misconceptions, making it difficult to measure true learning outcomes

7. BEHAVIOR

+ ITS INTENSITY [BE]
Students repeatedly practice concepts before examinations and interviews, often relying on textbooks, online videos, and mock interviews. They actively seek feedback to improve their understanding but rarely receive personalized evaluation. This is a high-intensity recurring behavior, especially during placement and examination seasons.

3. TRIGGERS TO ACT
[TR]

- PROS & CONS [AS]**
- Upcoming examinations
 - Technical interviews
 - Campus placement preparation
 - Poor performance in previous assessments
 - Need for self-assessment before learning new concepts
 - Recommendation from teachers or peers to evaluate conceptual understanding
 - Upcoming examinations
 - Technical interviews

10. YOUR SOLUTION
[SL]
Voice-Based Concept Understanding Analyser (ConceptCheckmate) is an AI-powered web application that enables students to explain concepts using their own voice. The system converts speech into text, performs Natural Language Processing (NLP) and semantic similarity analysis, identifies missing concepts and misconceptions, calculates a mastery score, and generates personalized feedback. This helps learners accurately evaluate their conceptual understanding and improve their knowledge before examinations and technical interviews.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- College Learning Management Systems (LMS)
- YouTube educational channels
- Coursera
- Udemy
- NPTEL
- ChatGPT and AI learning assistants

OFFLINE

- ☐ Classroom lectures
- ☐ Faculty guidance
- ☐ Library resources
- ☐ Peer group discussions
- ☐ Technical workshops
- ☐ Campus placement training sessions

4. EMOTIONS

BEFORE / AFTER [EM]

Before:

- **Confused**
- **Uncertain about conceptual understanding**
- **Anxious before exams or interviews**
- **Lack of confidence**

After:

- **Confident**
- **Motivated to improve**
- **Better understanding of weak areas**
- **Satisfied with personalized feedback and learning progress**